Linear Models II

Linear Algebra · Lecture 12

Last time (Lecture 11): if G is a g-inverse of $X'X$, then $\beta^0 = GX'y$ solves the normal equations, $H = XGX'$ gives the projection of y onto $\mathcal{C}(X)$ and does not depend on G, SSE $= y'My$, and $\hat{\sigma}^2 = \text{SSE}/(n-r)$ is unbiased (Theorems 10, 15). F1–F8: Chapters 1–6 of the textbook; F9, F10: probability, no proof. e is the vector of residuals; unit vectors are written as columns of I.

Facts

F1. $(AB)C = A(BC)$, $A(B+C) = AB + AC$, $(A+B)C = AC + BC$, $c(AB) = (cA)B = A(cB)$, $I_m A = A = AI_n$ $(A \in \mathbb{R}^{m\times n})$; $(A+B)' = A' + B'$, $(cA)' = cA'$, $(AB)' = B'A'$, $(A')' = A$; $x'y = y'x$; A is **symmetric** if $A' = A$.

F2. $(AB)_{ij} = \sum_r a_{ir}b_{rj}$. For $A = [a_1 \ \cdots \ a_p]$: $Ax = x_1a_1 + \cdots + x_pa_p$, $A\,\text{col}_j(I_p) = a_j$, $\text{col}_j(AB) = A\,\text{col}_j(B)$, $\text{row}_i(AB) = \text{row}_i(A)B$; if $Ax = Bx$ for all x, then $A = B$; $\text{diag}(\alpha_1,\ldots,\alpha_n)\,\text{diag}(\beta_1,\ldots,\beta_n) = \text{diag}(\alpha_1\beta_1,\ldots,\alpha_n\beta_n)$; block matrices multiply, and are transposed, block by block (when the blocks fit).

F3. $A \in \mathbb{R}^{n\times n}$ is invertible $\iff \operatorname{rank} A = n$; $BA = I \Rightarrow B = A^{-1}$; $(A')^{-1} = (A^{-1})'$, so the inverse of an invertible symmetric matrix is symmetric. $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible $\iff ad - bc \neq 0$, and then its inverse is $\frac{1}{ad-bc}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$.

F4. $\text{span}\{v_1,\ldots,v_k\} = \{c_1v_1 + \cdots + c_kv_k : c_1,\ldots,c_k \in \mathbb{R}\}$. For $A \in \mathbb{R}^{m\times n}$: $\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\}$ and $\mathcal{N}(A) = \{x : Ax = 0\}$ are subspaces; $\operatorname{rank} A = \dim \mathcal{C}(A) = \operatorname{rank} A' \leqslant \min\{m,n\}$; $\dim \mathcal{N}(A) = n - \operatorname{rank} A$. A has **full column rank** $(\operatorname{rank} A = n) \iff$ its columns are independent $\iff \mathcal{N}(A) = \{0\}$; **full row rank** $(\operatorname{rank} A = m) \iff \mathcal{C}(A) = \mathbb{R}^m$. $\operatorname{rank}(AB) \leqslant \operatorname{rank} A$. For subspaces $W_1 \subseteq W_2$, $\dim W_1 = \dim W_2 \Rightarrow W_1 = W_2$.

F5. $\operatorname{tr} A = \sum_i a_{ii}$; $\operatorname{tr}(A+B) = \operatorname{tr} A + \operatorname{tr} B$, $\operatorname{tr}(cA) = c\operatorname{tr} A$, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ $(A \in \mathbb{R}^{m\times n},\ B \in \mathbb{R}^{n\times m})$.

F6. $\|x\|^2 = x'x = x_1^2 + \cdots + x_n^2 > 0$ if $x \neq 0$; if $x \perp y$ $(x'y = 0)$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.

F7. $\mathbf{1} = \mathbf{1}_n = (1,\ldots,1)' \in \mathbb{R}^n$, $\mathbf{1}'\mathbf{1} = n$, $\bar{y} = \frac{1}{n}\mathbf{1}'y$, $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$: $\bar{J}_n' = \bar{J}_n = \bar{J}_n^2$, $\bar{J}_n y = \bar{y}\mathbf{1}$.

F8. For a subspace $W \subseteq \mathbb{R}^n$: $y \perp W$ if $y'w = 0$ for all $w \in W$; $W^\perp = \{y : y \perp W\}$ is a subspace, and $\dim W + \dim W^\perp = n$. For $A \in \mathbb{R}^{m\times n}$: $\mathcal{N}(A) = \mathcal{C}(A')^\perp$.

F9. (Probability.) $\mathsf{P}(z \notin \mathcal{A}) = 1 - \mathsf{P}(z \in \mathcal{A})$. $u_1,\ldots,u_m$ are **independent** if $\mathsf{P}(u_1 \in \mathcal{A}_1,\ldots,u_m \in \mathcal{A}_m) = \mathsf{P}(u_1 \in \mathcal{A}_1)\cdots\mathsf{P}(u_m \in \mathcal{A}_m)$ for all $\mathcal{A}_1,\ldots,\mathcal{A}_m$; then $g_1(u_1),\ldots,g_m(u_m)$ are independent, for any functions $g_1,\ldots,g_m$.

F10. Let $r,\nu,a,b \geqslant 1$ (the **degrees of freedom**). χ^2_r is the distribution of $w_1^2 + \cdots + w_r^2$ for independent $w_1,\ldots,w_r \sim N(0,1)$; t_ν is that of $Z/\sqrt{Y/\nu}$ for independent $Z \sim N(0,1)$, $Y \sim \chi^2_\nu$; $F_{a,b}$ is that of $(Y_1/a)/(Y_2/b)$ for independent $Y_1 \sim \chi^2_a$, $Y_2 \sim \chi^2_b$. For u with one of these distributions, $\mathsf{P}(u = c) = 0$ for every c; for $\tau \in (0,1)$, the τ **quantile** $t_{\nu,\tau}$, $F_{a,b,\tau}$ is the unique c with $\mathsf{P}(u \leqslant c) = \tau$; so $\mathsf{P}(u > c) = \alpha$ for the $(1-\alpha)$ quantile c.

From Lecture 8, Lecture 9 and Lecture 10 (restated).

- Lecture 8, Theorem 11: if $A \in \mathbb{R}^{n\times n}$ is symmetric and $A^2 = A$, then $\operatorname{rank} A = \operatorname{tr} A$.
- Lecture 8, Definition 13, Theorems 14(c), 15(b): a symmetric A is **positive definite** (PD) if $x'Ax > 0$ for all $x \neq 0$; a PD matrix is invertible; for $A \in \mathbb{R}^{n\times p}$, $A'A$ is PD $\iff \operatorname{rank} A = p$.
- Lecture 9, Definitions 1, 7, Theorem 3(b): for a random vector z, $\mathsf{E}(z) = (\mathsf{E}(z_i))$, $\text{var}(z) = \mathsf{E}\big((z-\mu)(z-\mu)'\big)$ $(\mu = \mathsf{E}(z))$. An **estimator** of a parameter θ is a random variable $T = g(y)$, g involving no unknown parameter; it is **unbiased** if $\mathsf{E}(T) = \theta$ for all values of the parameters. For $\mathsf{E}(z) = \mu$, $\text{var}(z) = \Sigma$ and constant A, b: $\mathsf{E}(Az+b) = A\mu + b$, $\text{var}(Az+b) = A\Sigma A'$.
- Lecture 10, Definition 1, Proposition 2, Theorem 3: $N_n(\mu,\Sigma)$ $(\mu \in \mathbb{R}^n$, Σ symmetric with $x'\Sigma x \geqslant 0$ for all $x)$ is the **multivariate normal** distribution, $N(0,1)$ the standard normal. If $z \sim N_n(\mu,\Sigma)$: $\mathsf{E}(z) = \mu$, $\text{var}(z) = \Sigma$, and $Az + b \sim N_m(A\mu + b,\ A\Sigma A')$ for constant $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$.

- Lecture 10, Theorem 12: if $T \sim t_\nu$ and $\alpha \in (0,1)$, then $\mathsf{P}\big(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}\big) = 1-\alpha$ and $t_{\nu,1-\alpha/2} > 0$.
- Lecture 10, Theorems 15, 16(a),(b): let $z \sim N_n(\mu, \sigma^2 I_n)$ ($\sigma > 0$), and $A, B \in \mathbb{R}^{n\times n}$ symmetric with $A^2 = A$, $B^2 = B$. If $\operatorname{rank} A = r \geqslant 1$ and $A\mu = 0$, then $z'Az/\sigma^2 \sim \chi^2_r$. If $AB = 0$, then $z'Az$ and $z'Bz$ are independent. If $K \in \mathbb{R}^{m\times n}$ and $KA = 0$, then Kz and $z'Az$ are independent.
- Lecture 10, Definition 18: if the distribution of the observations depends on parameters and $\alpha \in (0,1)$, an interval $[L, U]$ whose endpoints $L \leqslant U$ are computed from the observations is a **confidence interval** with level $1-\alpha$ for a parameter $\theta \in \mathbb{R}$ if $\mathsf{P}(L \leqslant \theta \leqslant U) = 1-\alpha$ for all values of the parameters.

From Lecture 11 (restated; $X \in \mathbb{R}^{n\times p}$, $y \in \mathbb{R}^n$).

- Definition 1 and the lines after it: in the **linear model** $y = X\beta + \varepsilon$, the design matrix X is a known constant matrix, $\beta \in \mathbb{R}^p$ an unknown constant vector of parameters, the errors $\varepsilon \in \mathbb{R}^n$ a random vector, and the response y is observed. (A1) $\mathsf{E}(y) = X\beta$; (A2) $\mathsf{E}(\varepsilon) = 0$; (A3) $\operatorname{var}(\varepsilon) = \sigma^2 I_n$ ($\sigma^2 > 0$); (A4) $\operatorname{rank} X = p$. (A1) $\iff$ (A2), and (A3) gives $\operatorname{var}(y) = \sigma^2 I_n$. A **least-squares estimate** is a $\hat\beta \in \mathbb{R}^p$ that minimizes $S(\beta) = \|y - X\beta\|^2$; $X'X\beta = X'y$ are the **normal equations**.
- Theorem 2: (a) the normal equations have a solution; (b) $X'X$ is invertible $\iff$ $\operatorname{rank} X = p$. Theorem 3: (a) $\hat\beta$ is a least-squares estimate $\iff$ $X'X\hat\beta = X'y$. (b) All least-squares estimates give the same $\hat y = X\hat\beta$ (the **fitted values**), the orthogonal projection of y onto $\mathcal{C}(X)$ (the $\hat w \in \mathcal{C}(X)$ with $y - \hat w \perp \mathcal{C}(X)$); $e = y - \hat y$. (c) If $\operatorname{rank} X = p$, $\hat\beta = (X'X)^{-1}X'y$ is the only least-squares estimate.
- Theorem 6: n observations in g groups of sizes $n_1, \dots, n_g \geqslant 1$ (first the n_1 of group 1, and so on); the **indicator** $d_l \in \mathbb{R}^n$ of group l has entry 1 in the rows of group l, 0 elsewhere. The model $y_i = \mu + \alpha_l + \varepsilon_i$ (observation i in group l) is $y = X\beta + \varepsilon$, $\beta = (\mu, \alpha_1, \dots, \alpha_g)'$, $X = [\mathbf{1}_n \; d_1 \; \cdots \; d_g]$. For $X_1 = [d_1 \; \cdots \; d_g]$: (a) X_1 has full column rank; (b) $\operatorname{rank} X = g$, $\mathcal{N}(X) = \operatorname{span}\{(-1, 1, \dots, 1)'\}$; (c) $\mathcal{C}(X_1) = \mathcal{C}(X)$.
- Example 7: $y = (5,7,6,3,5,2)'$, $g = 3$, $(n_1, n_2, n_3) = (3,2,1)$; $X = \begin{pmatrix} 1&1&0&0\\ 1&1&0&0\\ 1&1&0&0\\ 1&0&1&0\\ 1&0&1&0\\ 1&0&0&1 \end{pmatrix}$, $X'X = \begin{pmatrix} 6&3&2&1\\ 3&3&0&0\\ 2&0&2&0\\ 1&0&0&1 \end{pmatrix}$, $X'y = (28, 18, 8, 2)'$; the least-squares estimates are $(-t, 6+t, 4+t, 2+t)'$ ($t \in \mathbb{R}$), and $\hat y = (6,6,6,4,4,2)'$.
- Definition 8: G with $AGA = A$ is a **g-inverse** of A; if A is invertible, A^{-1} is its only g-inverse.
- Theorem 10: let $r = \operatorname{rank} X$, G a g-inverse of $X'X$, $H = XGX'$ (the **hat matrix**) and $M = I_n - H$. (a) $XGX'X = X$, $X'XGX' = X'$; $\beta^0 = GX'y$ solves the normal equations. (b) Hy is the orthogonal projection of y onto $\mathcal{C}(X)$, so H does not depend on the choice of G; $H = X(X'X)^{-1}X'$ if $r = p$. (c) $H' = H = H^2$, $HX = X$, $X'H = X'$; $M' = M = M^2$, $MX = 0$, $HM = MH = 0$. (d) $\operatorname{rank} H = \operatorname{tr} H = r$, $\operatorname{rank} M = \operatorname{tr} M = n - r$. (e) For every least-squares estimate $\hat\beta$: $\hat y = Hy$, $e = My$, and $\mathrm{SSE} = S(\hat\beta) = e'e = y'My = y'y - \hat\beta' X'y$.
- Example 11: in Example 7, $G = \operatorname{diag}(0, \frac13, \frac12, 1)$ is a g-inverse of $X'X$, $\beta^0 = (0,6,4,2)'$, $e = (-1,1,0,-1,1,0)'$, $\mathrm{SSE} = 4$.
- Theorem 13 (Gauss–Markov): under (A1)–(A4), for $\hat\beta = (X'X)^{-1}X'y$ and every $\tilde\beta = Ly$ with $\mathsf{E}(Ly) = \beta$ for all β: $\operatorname{var}(a'\tilde\beta) \geqslant \operatorname{var}(a'\hat\beta)$ for every $a \in \mathbb{R}^p$.
- Theorem 15 and the line after it: under (A2), (A3) and $r = \operatorname{rank} X < n$, $\mathsf{E}(\mathrm{SSE}) = (n-r)\sigma^2$, so $\hat\sigma^2 = \mathrm{SSE}/(n-r)$ is unbiased for σ^2; in Example 11, $\hat\sigma^2 = 4/(6-3) = \frac43$.
- Definition 16, Theorem 17: $\mathrm{SST} = \sum_i (y_i - \bar y)^2$, $\mathrm{SSR} = \sum_i (\hat y_i - \bar y)^2$, $\mathrm{SSE} = \sum_i (y_i - \hat y_i)^2$. If $\mathbf{1}_n \in \mathcal{C}(X)$: $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ and $y'y = n\bar y^2 + \mathrm{SSR} + \mathrm{SSE}$.

1. Estimable functions and their best linear unbiased estimators

Definition 1. Assume (A1), with $X \in \mathbb{R}^{n\times p}$ of any rank, and let $a \in \mathbb{R}^p$. A **linear estimator** is a random variable $c'y$ with a constant $c \in \mathbb{R}^n$; it is a **linear unbiased estimator** of $a'\beta$ if $\mathsf{E}(c'y) = a'\beta$ for every $\beta \in \mathbb{R}^p$. $a'\beta$ is **estimable** if it has a linear unbiased estimator. A linear unbiased estimator of $a'\beta$ with the smallest variance among all of them is a **best linear unbiased estimator** (**BLUE**) of $a'\beta$.

Theorem 2. Assume (A1), and let $a \in \mathbb{R}^p$.

(a) A linear estimator $c'y$ is unbiased for $a'\beta \iff X'c = a$. Hence $a'\beta$ is estimable $\iff a \in \mathcal{C}(X') \iff a \perp \mathcal{N}(X)$. If $\operatorname{rank} X = p$, every $a'\beta$ is estimable.

(b) Let G be a g-inverse of $X'X$ and $H = XGX'$. If $a = X'c$, then $a'\hat\beta = c'\hat y = c'Hy$ for every least-squares estimate $\hat\beta$ (the same linear estimator for all of them), and $\mathsf{E}(a'\hat\beta) = a'\beta$.

(c) If $a'\beta$ is not estimable, then $a'\hat\beta$ takes every real value as $\hat\beta$ runs through the least-squares estimates.

Proof.

(a) Lecture 9, Theorem 3(b), and (A1); Definition 1:

$$\mathsf{E}(c'y) = c'X\beta; \qquad c'X\beta = a'\beta \text{ for all } \beta \implies c'X\operatorname{col}_j(I_p) = a'\operatorname{col}_j(I_p)\ (j = 1,\dots,p)$$
$$\implies c'X = a',\ X'c = a; \qquad \text{conversely } X'c = a \implies \mathsf{E}(c'y) = c'X\beta = a'\beta \qquad \text{(F2, F1)}$$
$$\implies \Big(a'\beta \text{ estimable} \iff a \in \mathcal{C}(X')\Big) \qquad \text{(F4)}$$
$$\mathcal{N}(X) = \mathcal{C}(X')^\perp \implies \mathcal{C}(X') \subseteq \mathcal{N}(X)^\perp \qquad \text{(F8)}$$
$$\dim\mathcal{C}(X') = \operatorname{rank}X' = \operatorname{rank}X = p - \dim\mathcal{N}(X) = \dim\mathcal{N}(X)^\perp \implies \mathcal{C}(X') = \mathcal{N}(X)^\perp \qquad \text{(F4, F8)}$$
$$\operatorname{rank}X = p \implies \operatorname{rank}X' = p \implies \mathcal{C}(X') = \mathbb{R}^p \qquad \text{(F4)}$$

(b) $a = X'c$; Lecture 9, Theorem 3(b), and (A1) for the mean:

$$X\hat\beta = \hat y = Hy \implies a'\hat\beta = c'X\hat\beta = c'\hat y = c'Hy \qquad \text{(Lecture 11, Theorems 3(b), 10(e))}$$
$$\mathsf{E}(c'Hy) = c'HX\beta = c'X\beta = a'\beta \qquad \text{(Lecture 11, Theorem 10(c))}$$

(c) By (a), $a \notin \mathcal{N}(X)^\perp$; a least-squares estimate $\hat\beta$ exists (Lecture 11, Theorems 2(a), 3(a)):

$$a'z \neq 0 \text{ for some } z \in \mathcal{N}(X)$$
$$Xz = 0 \implies X'X(\hat\beta + tz) = X'X\hat\beta = X'y \quad (t \in \mathbb{R})$$
$$\implies \hat\beta + tz \text{ is a least-squares estimate} \qquad \text{(Lecture 11, Theorem 3(a))}$$
$$a'(\hat\beta + tz) = a'\hat\beta + t\,a'z \text{ takes every real value}$$

□

Example 3. In Lecture 11, Example 7, which of $\mu+\alpha_1$, $\alpha_1-\alpha_2$, μ, α_1 are estimable? With the group means $\bar y_1 = \frac{1}{3}(y_1+y_2+y_3)$, $\bar y_2 = \frac{1}{2}(y_4+y_5)$, $\bar y_3 = y_6$, find the estimator of $\alpha_1-\alpha_2$ of Theorem 2(b).

Solution.

$\mathcal{N}(X) = \operatorname{span}\{(-1,1,1,1)'\}$ (Lecture 11, Theorem 6(b)); Theorem 2(a),(c):

$a_0\mu + a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$ is estimable $\iff -a_0 + a_1 + a_2 + a_3 = 0$

$\mu+\alpha_1:\ -1+1=0, \quad \alpha_1-\alpha_2:\ 1-1=0 \implies$ estimable; on $(-t, 6+t, 4+t, 2+t)'$: values 6, 2

$\mu:\ -1 \neq 0, \quad \alpha_1:\ 1 \neq 0 \implies$ not estimable; values $-t,\ 6+t$ change with t

$X'y$ holds the total and the group totals (F2); Lecture 11, Theorem 3(a):

$$X'X(0, \bar{y}_1, \bar{y}_2, \bar{y}_3)' = (3\bar{y}_1 + 2\bar{y}_2 + \bar{y}_3,\ 3\bar{y}_1,\ 2\bar{y}_2,\ \bar{y}_3)' = X'y$$
$$\hat{\beta} = (0, \bar{y}_1, \bar{y}_2, \bar{y}_3)' \text{ is a least-squares estimate}$$
$$\hat{\alpha}_1 - \hat{\alpha}_2 = \bar{y}_1 - \bar{y}_2 = c'y, \qquad c = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -\tfrac{1}{2}, -\tfrac{1}{2}, 0\right)'$$
$$X'c = (1-1,\ 1,\ -1,\ 0)' = (0, 1, -1, 0)'$$

Theorem 4. Assume (A1)–(A3), with X of any rank; let G be a g-inverse of $X'X$, $H = XGX'$, $M = I_n - H$, and let $a'\beta$ be estimable. Then $a'\hat{\beta}$, the same for every least-squares estimate $\hat{\beta}$, is the BLUE of $a'\beta$: it is a linear unbiased estimator of $a'\beta$, and every linear unbiased estimator $c'y$ of $a'\beta$ has $\mathrm{var}(c'y) \geqslant \mathrm{var}(a'\hat{\beta})$, with equality only if $c'y = a'\hat{\beta}$ for every y. Moreover $\mathrm{var}(a'\hat{\beta}) = \sigma^2 a'Ga$ (for every such G).

Proof.

$c'y$ a linear unbiased estimator of $a'\beta$; $\mathrm{var}(y) = \sigma^2 I_n$ ((A3)) and Lecture 9, Theorem 3(b) for the variances; $H' = H = H^2$, $M' = M = M^2$ (Lecture 11, Theorem 10(c)); $u = Mc$:

$$X'c = a, \quad a'\hat{\beta} = c'Hy \text{ is linear unbiased} \qquad \text{(Theorem 2(a),(b))}$$
$$\mathrm{var}(c'y) = \sigma^2 c'c, \quad \mathrm{var}(c'Hy) = \sigma^2 c'HH'c = \sigma^2 c'Hc$$
$$\mathrm{var}(c'y) - \mathrm{var}(a'\hat{\beta}) = \sigma^2 c'(I_n - H)c = \sigma^2 c'M'Mc = \sigma^2\|u\|^2 \geqslant 0 \qquad \text{(F6)}$$
$$\sigma^2\|u\|^2 = 0 \implies u = 0 \implies c = Hc \implies c'y = c'Hy = a'\hat{\beta} \text{ (all } y) \qquad \text{(F6)}$$
$$c'Hc = c'XGX'c = a'Ga$$

□

So $\mathrm{var}(c'y) = \mathrm{var}(a'\hat{\beta}) + \sigma^2\|Mc\|^2$. Under (A4), every $a'\beta$ is estimable (Theorem 2(a)), and Theorem 4 is Lecture 11, Theorem 13 for one $a'\beta$ at a time.

Example 5. In Lecture 11, Examples 7 and 11 ($G = \mathrm{diag}(0, \frac{1}{3}, \frac{1}{2}, 1)$), under (A1), (A3): find $\beta^0 = GX'y$, $\mathsf{E}(\beta^0)$, $\mathrm{var}(\beta^0)$ and $\mathrm{var}(\bar{y}_1 - \bar{y}_2)$.

Solution.

$GX'X$: the rows of $X'X$ times $0, \frac{1}{3}, \frac{1}{2}, 1$; $X'y$ as in Example 3; Lecture 9, Theorem 3(b):

$$GX'X = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \beta^0 = GX'y = (0, \bar{y}_1, \bar{y}_2, \bar{y}_3)'$$
$$\mathsf{E}(\beta^0) = GX'X\beta = (0,\ \mu + \alpha_1,\ \mu + \alpha_2,\ \mu + \alpha_3)' \qquad \text{((A1))}$$
$$\mathrm{var}(\beta^0) = GX'(\sigma^2 I_n)XG' = \sigma^2 GX'XG = \sigma^2\,\mathrm{diag}(0, \tfrac{1}{3}, \tfrac{1}{2}, 1) \qquad \text{((A3); } G' = G\text{; F2)}$$
$$\mathrm{var}(\bar{y}_1 - \bar{y}_2) = a'\,\mathrm{var}(\beta^0)a = \sigma^2\left(\tfrac{1}{3} + \tfrac{1}{2}\right) = \tfrac{5}{6}\sigma^2 = \sigma^2 a'Ga \qquad (a = (0, 1, -1, 0)')$$

So β^0 is the $\hat{\beta}$ of Example 3, and $\bar{y}_l$ estimates $\mu + \alpha_l$, not α_l; $GX'XG$ has the columns of $GX'X$ times $0, \frac{1}{3}, \frac{1}{2}, 1$; and $\frac{5}{6}\sigma^2$ is the variance of Theorem 4.

2. The normal linear model; t intervals and tests

Definition 6. The **normal linear model** is the linear model of Lecture 11, Definition 1, with $X \in \mathbb{R}^{n\times p}$ of any rank, under (A5) $\varepsilon \sim N_n(0, \sigma^2 I_n)$ for a number $\sigma^2 > 0$. Then (A2), (A3) hold (Lecture 10, Proposition 2), and $y = X\beta + I_n\varepsilon \sim N_n(X\beta, \sigma^2 I_n)$ (Lecture 10, Theorem 3), so (A1) holds. Let $\alpha \in (0,1)$. A **hypothesis** H_0 is a condition on the parameters, such as $K\beta = m$. A **test** of H_0 is given by a **rejection region** $\mathcal{R}$, a set of possible values of y: H_0 is rejected if the observed y lies in $\mathcal{R}$. The test has **level** α if $\mathsf{P}(y \in \mathcal{R}) = \alpha$ for all parameter values that satisfy H_0. Usually $\mathcal{R}$ is described by a **test statistic** T computed from y, as $\{T > c\}$ or $\{|T| > c\}$. When H_0 is rejected at level α, the departure from H_0 is **significant** at level α. Confidence intervals are those of Lecture 10, Definition 18; the quantiles $t_{\nu,\tau}$, $F_{a,b,\tau}$ those of F10.

Theorem 7. In the normal linear model let $\operatorname{rank} X = r < n$, G a g-inverse of $X'X$, $\beta^0 = GX'y$, $H = XGX'$, $M = I_n - H$, and let $K \in \mathbb{R}^{q\times p}$ be such that every entry of $K\beta$ is estimable.

(a) $K\beta^0 \sim N_q(K\beta, \sigma^2 KGK')$, and $K\beta^0$, KGK' are the same for every g-inverse G of $X'X$. Under (A4), $\hat\beta = (X'X)^{-1}X'y \sim N_p(\beta, \sigma^2(X'X)^{-1})$.

(b) $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}$, and $\mathsf{P}(\mathrm{SSE} = 0) = 0$.

(c) $K\beta^0$ and SSE are independent.

Proof.

(a) Row i of K is $t_i'X$, $t_i \in \mathbb{R}^n$ (Theorem 2(a)); $T \in \mathbb{R}^{q\times n}$ has the rows t_i', so $K = TX$ (F2); $y \sim N_n(X\beta, \sigma^2 I_n)$ (Definition 6):

$$K\beta^0 = TXGX'y = THy \sim N_q(THX\beta,\ \sigma^2 THH'T') \qquad \text{(Lecture 10, Theorem 3)}$$
$$THX\beta = TX\beta = K\beta, \qquad THH'T' = THT' = TXGX'T' = KGK' \qquad \text{(Lecture 11, Theorem 10(c))}$$

H does not depend on G (Lecture 11, Theorem 10(b)), so neither do $K\beta^0 = THy$ and $KGK' = THT'$. Under (A4), $X'X$ is invertible and $(X'X)^{-1}$ is its only g-inverse (Lecture 11, Theorem 2(b), Definition 8), every $a'\beta$ is estimable (Theorem 2(a)), and $K = I_p$ gives $\hat\beta$.

(b) $y \sim N_n(X\beta, \sigma^2 I_n)$; Lecture 11, Theorem 10(c)–(e):

$$\mathrm{SSE} = y'My, \quad M' = M = M^2$$
$$\operatorname{rank} M = n - r \geqslant 1, \quad MX\beta = 0$$
$$\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r} \qquad \text{(Lecture 10, Theorem 15, } z = y,\ A = M\text{)}$$
$$\mathsf{P}(\mathrm{SSE} = 0) = \mathsf{P}(\mathrm{SSE}/\sigma^2 = 0) = 0 \qquad \text{(F10)}$$

(c) $K\beta^0 = (TH)y$ ((a)); Lecture 10, Theorem 16(b), with TH, M for its K, A:

$$(TH)M = T(HM) = 0 \implies K\beta^0 \text{ and } \mathrm{SSE} = y'My \text{ independent} \qquad \text{(Lecture 11, Theorem 10(c))}$$

□

We write $s^2 = \hat\sigma^2 = \mathrm{SSE}/(n-r)$ (unbiased by Lecture 11, Theorem 15) and $s = \sqrt{s^2}$; by (b), $s > 0$ with probability 1.

Theorem 8. In the normal linear model let $\operatorname{rank} X = r < n$, G a g-inverse of $X'X$, $\beta^0 = GX'y$, and let $a \in \mathbb{R}^p$ be nonzero with $a'\beta$ estimable. Then $a'Ga > 0$; let $\operatorname{SE}(a'\beta^0) = s\sqrt{a'Ga}$, the **standard error** of $a'\beta^0$, and $t^* = t_{n-r,1-\alpha/2}$.

(a) $T = (a'\beta^0 - a'\beta)/\operatorname{SE}(a'\beta^0) \sim t_{n-r}$.

(b) $\left[a'\beta^0 - t^*\operatorname{SE}(a'\beta^0),\ a'\beta^0 + t^*\operatorname{SE}(a'\beta^0)\right]$ is a confidence interval of level $1-\alpha$ for $a'\beta$. For $m_0 \in \mathbb{R}$, the test of $H_0\colon a'\beta = m_0$ that rejects when $|T_0| > t^*$, where $T_0 = (a'\beta^0 - m_0)/\operatorname{SE}(a'\beta^0)$, has level α; it rejects exactly when m_0 lies outside this interval.

$\operatorname{SE}(a'\beta^0)$ and the interval are the same for every g-inverse G of $X'X$ (Theorem 7(a), $K = a'$; s by Lecture 11, Theorem 10(e)). Under (A4), $G = (X'X)^{-1}$, $\beta^0 = \hat\beta$ and $r = p$.

Proof.

$a'Ga > 0$: $a = X'c$ (Theorem 2(a)); $H = XGX'$:

$$a'Ga = c'XGX'c = c'Hc = c'H'Hc = \|Hc\|^2 \qquad \text{(Lecture 11, Theorem 10(c))}$$
$$X'Hc = X'c = a \neq 0 \implies Hc \neq 0 \implies a'Ga > 0 \qquad \text{(F6)}$$

(a) Lecture 10, Theorem 3 for z; $s > 0$ with probability 1:

$$z = \frac{a'\beta^0 - a'\beta}{\sigma\sqrt{a'Ga}} \sim N(0,1) \qquad \text{(Theorem 7(a), } K = a')$$
$$v = \operatorname{SSE}/\sigma^2 \sim \chi^2_{n-r}; \quad z,\ v \text{ independent} \qquad \text{(Theorem 7(b),(c); F9)}$$
$$T = \frac{z}{s/\sigma} = \frac{z}{\sqrt{v/(n-r)}} \sim t_{n-r} \qquad \text{(F10)}$$

(b) $t^* > 0$; $\operatorname{SE} = \operatorname{SE}(a'\beta^0) > 0$ with probability 1:

$$\mathsf{P}(|T| \leqslant t^*) = 1 - \alpha \qquad \text{((a); Lecture 10, Theorem 12)}$$
$$|T| \leqslant t^* \iff a'\beta^0 - t^*\operatorname{SE} \leqslant a'\beta \leqslant a'\beta^0 + t^*\operatorname{SE} \qquad \text{(Lecture 10, Definition 18)}$$
$$H_0\colon a'\beta = m_0 \implies T_0 = T$$
$$\mathsf{P}(|T_0| > t^*) = 1 - (1-\alpha) = \alpha \text{ under } H_0 \qquad \text{(F9)}$$
$$|T_0| > t^* \iff |a'\beta^0 - m_0| > t^*\operatorname{SE} \iff m_0 \text{ outside the interval}$$

□

Example 9. In Lecture 11, Example 7, assume the normal linear model. Find the confidence interval of level $1-\alpha$ for $\alpha_1 - \alpha_2$, and the test of $H_0\colon \alpha_1 = \alpha_2$ at level α.

Solution.

$\alpha_1 - \alpha_2$ is estimable (Example 3); Example 5, Lecture 11, Example 11 and Theorem 15; Theorem 8(b):

$$a = (0,1,-1,0)'\colon \quad a'\beta^0 = 6 - 4 = 2, \quad a'Ga = \tfrac{1}{3} + \tfrac{1}{2} = \tfrac{5}{6}; \quad \operatorname{SSE} = 4,\ n - r = 6 - 3 = 3,\ s^2 = \tfrac{4}{3}$$
$$\operatorname{SE} = s\sqrt{5/6} = \sqrt{10/9} = \sqrt{10}/3; \quad \text{interval } \left[2 - t_{3,1-\alpha/2}\sqrt{10}/3,\ 2 + t_{3,1-\alpha/2}\sqrt{10}/3\right]$$
$$m_0 = 0\colon \quad T_0 = 2/(\sqrt{10}/3) = 6/\sqrt{10}; \quad \text{the test rejects } H_0 \iff t_{3,1-\alpha/2} < 6/\sqrt{10}$$

Equivalently, the interval contains 0 exactly when $t_{3,1-\alpha/2} \geqslant 6/\sqrt{10}$.

3. Testable hypotheses and the F test

Definition 10. Let $\operatorname{rank} X = r$, $K \in \mathbb{R}^{q\times p}$ and $m \in \mathbb{R}^q$. The hypothesis $H_0\colon K\beta = m$ is **testable** if every entry of $K\beta$ is estimable and $\operatorname{rank} K = q$ (the rows of K are linearly independent).

Let $H_0\colon K\beta = m$ be testable, G a g-inverse of $X'X$, $H = XGX'$; $K = T_1X$, $T_1 \in \mathbb{R}^{q\times n}$ (Theorem 2(a), F2):

$$
\begin{aligned}
&T = T_1H: \quad TX = T_1HX = T_1X = K, \quad TH = T_1H^2 = T && \text{(Lecture 11, Theorem 10(c))}\\
&q = \operatorname{rank}(TX) \leqslant \operatorname{rank} T \leqslant q \implies \operatorname{rank} T' = \operatorname{rank} T = q && \text{(F4)}\\
&V = KGK' = TXGX'T' = THT' = THH'T' = TT' && \text{(Lecture 11, Theorem 10(c))}\\
&\quad\implies V \text{ is PD } (A = T'), \text{ hence invertible} && \text{(Lecture 8, Theorems 15(b), 14(c))}
\end{aligned}
$$

$T = T_1H$ and $V = THT'$ do not depend on G (Lecture 11, Theorem 10(b)). Under (A4), $H_0\colon K\beta = m$ is testable for every K of rank q (Theorem 2(a)).

Theorem 11. In the normal linear model let $\operatorname{rank} X = r < n$, G a g-inverse of $X'X$, $\beta^0 = GX'y$, and $H_0\colon K\beta = m$ testable, with $K \in \mathbb{R}^{q\times p}$ and $V = KGK'$.

(a) Let $F = \Big[(K\beta^0 - m)'V^{-1}(K\beta^0 - m)/q\Big]/s^2$, defined with probability 1 (V invertible by the lines after Definition 10, $s^2 > 0$ by Theorem 7(b)) and the same for every g-inverse G of $X'X$. If H_0 holds, then $F \sim F_{q,n-r}$; so the test that rejects H_0 when $F > F_{q,n-r,1-\alpha}$ has level α.

(b) If $q = 1$ and $K = a'$, then $F = T_0^2$, with T_0 of Theorem 8(b) for $m_0 = m$.

Proof. (a) $K\beta^0$ and V do not depend on G (Theorem 7(a), the lines after Definition 10), nor does $s^2 = y'My/(n-r)$ (Lecture 11, Theorem 10(b),(e)).

Algebra: T as after Definition 10 ($TH = T$, $V = TT'$); H_0 holds; $\varepsilon = y - X\beta$; $MX = 0$, $X'M = 0$ (Lecture 11, Theorem 10(c)):

$$
\begin{aligned}
K\beta^0 &= TXGX'y = THy = Ty\\
K\beta^0 - m &= Ty - TX\beta = T\varepsilon && (m = K\beta = TX\beta)\\
U = T'V^{-1}T:\ U' &= U, \quad U^2 = T'V^{-1}(TT')V^{-1}T = U && (V^{-1} \text{ symmetric, F3})\\
\operatorname{rank} U &= \operatorname{tr} U = \operatorname{tr}(V^{-1}TT') = \operatorname{tr} I_q = q && \text{(Lecture 8, Theorem 11; F5)}\\
TM &= T - TH = 0 \implies UM = 0\\
Q &= (K\beta^0 - m)'V^{-1}(K\beta^0 - m) = \varepsilon' U\varepsilon\\
\mathrm{SSE} &= y'My = \varepsilon' M\varepsilon
\end{aligned}
$$

Distribution: $\varepsilon \sim N_n(0, \sigma^2 I_n)$ (A5); $M' = M = M^2$, $\operatorname{rank} M = n - r$ (Lecture 11, Theorem 10(c),(d)):

$$
\begin{aligned}
Q/\sigma^2 = \varepsilon' U\varepsilon/\sigma^2 &\sim \chi^2_q && \text{(Lecture 10, Theorem 15, } \mu = 0)\\
\mathrm{SSE}/\sigma^2 = \varepsilon' M\varepsilon/\sigma^2 &\sim \chi^2_{n-r} && \text{(Lecture 10, Theorem 15, } \mu = 0)\\
Q \text{ and SSE} &\text{ independent} && \text{(Lecture 10, Theorem 16(a), } UM = 0)\\
F = \frac{(Q/\sigma^2)/q}{(\mathrm{SSE}/\sigma^2)/(n-r)} &\sim F_{q,n-r} && \text{(F9; F10)}\\
\mathsf{P}(F > F_{q,n-r,1-\alpha}) &= \alpha && \text{(F10)}
\end{aligned}
$$

(b) $K = a'$, $V = a'Ga$, $\mathrm{SE}(a'\beta^0)^2 = s^2 a'Ga$ (Theorem 8):

$$
F = \frac{(a'\beta^0 - m)^2}{s^2 a'Ga} = \left(\frac{a'\beta^0 - m}{\mathrm{SE}(a'\beta^0)}\right)^2 = T_0^2
$$

□

4. One-way analysis of variance

Theorem 12. In the one-way design of Lecture 11, Theorem 6, let $\mathcal{I}_l$ be the set of the observations (row indices) of group l, $\bar{y}_l = n_l^{-1}\sum_{i\in\mathcal{I}_l} y_i$ the **group mean** and $\bar{y}$ the **grand mean** of all n observations. Then $\hat{y} = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g$ (the fitted value of each observation is the mean of its group), and

$$\sum_{i=1}^{n}(y_i-\bar{y})^2 = \sum_{l=1}^{g} n_l(\bar{y}_l-\bar{y})^2 + \sum_{l=1}^{g}\sum_{i\in\mathcal{I}_l}(y_i-\bar{y}_l)^2$$

that is, SST = SSR + SSE, with the **between-group** sum of squares SSR and the **within-group** sum of squares SSE.

Proof.

Fitted values: X_1 has full column rank (Lecture 11, Theorem 6(a)); $X_1'y$ holds the group totals $d_l'y$ (F2); no observation lies in two groups:

$$d_l'd_m = 0\ (l\neq m),\quad d_l'd_l = n_l \implies X_1'X_1 = \operatorname{diag}(n_1,\ldots,n_g) \quad \text{(F2)}$$
$$\hat{\beta}_1 = (X_1'X_1)^{-1}X_1'y = (\bar{y}_1,\ldots,\bar{y}_g)' \quad \text{(F3; Lecture 11, Theorem 3(c))}$$
$$X_1\hat{\beta}_1 = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g = \hat{y} \quad (\mathcal{C}(X_1)=\mathcal{C}(X)) \quad \text{(Lecture 11, Theorems 6(c), 3(b))}$$

Sums of squares: $\hat{y}_i = \bar{y}_l$ for $i\in\mathcal{I}_l$:

$$\text{SSR} = \textstyle\sum_i(\hat{y}_i-\bar{y})^2 = \sum_l n_l(\bar{y}_l-\bar{y})^2$$
$$\text{SSE} = \textstyle\sum_i(y_i-\hat{y}_i)^2 = \sum_l\sum_{i\in\mathcal{I}_l}(y_i-\bar{y}_l)^2 \quad \text{(Lecture 11, Definition 16)}$$
$$\text{SST} = \text{SSR} + \text{SSE} \quad (\mathbf{1}_n = \textstyle\sum_l d_l \in \mathcal{C}(X)) \quad \text{(Lecture 11, Theorem 17)}$$

□

Corollary 13. In the one-way design of Lecture 11, Theorem 6, use the design $X_1 = [d_1\ \cdots\ d_g]$ with $\beta = (\mu_1,\ldots,\mu_g)'$ (the group means), and assume the normal linear model with $n > g \geqslant 2$. With SSR, SSE of Theorem 12, let $F = \big[\text{SSR}/(g-1)\big]/\big[\text{SSE}/(n-g)\big]$, defined with probability 1 (Theorem 7(b)). If $H_0\colon \mu_1 = \cdots = \mu_g$ holds, then $F \sim F_{g-1,n-g}$; so the test that rejects H_0 when $F > F_{g-1,n-g,1-\alpha}$ has level α.

Proof.

$\operatorname{rank} X_1 = g$ (Lecture 11, Theorem 6(a)), so $H = X_1(X_1'X_1)^{-1}X_1'$ and $M = I_n - H$ have $H' = H = H^2$, $HX_1 = X_1$, $HM = 0$, $\operatorname{tr} H = g$, $Hy = \hat{y}$ (Lecture 11, Theorems 2(b), 10, design X_1); $\mathbf{1}_n = X_1\mathbf{1}_g$; $A = H - \bar{J}_n$:

$$H\bar{J}_n = \tfrac{1}{n}(HX_1\mathbf{1}_g)\mathbf{1}' = \tfrac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n,\quad \bar{J}_nH = (H\bar{J}_n)' = \bar{J}_n \quad \text{(F1, F7)}$$
$$A' = A,\quad A^2 = H - 2\bar{J}_n + \bar{J}_n^2 = A \quad \text{(F7)}$$
$$\operatorname{rank} A = \operatorname{tr} A = g - \tfrac{1}{n}\operatorname{tr}(\mathbf{1}\mathbf{1}') = g - \tfrac{1}{n}\mathbf{1}'\mathbf{1} = g-1 \quad \text{(Lecture 8, Theorem 11; F5)}$$
$$AM = HM - \bar{J}_n(I_n - H) = -\bar{J}_n + \bar{J}_n = 0$$
$$y'Ay = y'A'Ay = \|Hy - \bar{J}_n y\|^2 = \|\hat{y} - \bar{y}\mathbf{1}_n\|^2 = \text{SSR} \quad \text{(F7; Theorem 12)}$$
$$y'My = e'e = \text{SSE} \quad \text{(Lecture 11, Theorem 10(e))}$$

Distribution: $y \sim N_n(X_1\beta, \sigma^2 I_n)$ (Definition 6); $g - 1 \geqslant 1$:

$$AX_1\beta = \mu_1 A(d_1+\cdots+d_g) = \mu_1(H\mathbf{1}_n - \bar{J}_n\mathbf{1}_n) = \mu_1(\mathbf{1}_n - \mathbf{1}_n) = 0 \quad (H_0;\ \text{F7})$$
$$\text{SSR}/\sigma^2 = y'Ay/\sigma^2 \sim \chi^2_{g-1} \quad \text{(Lecture 10, Theorem 15)}$$
$$\text{SSE}/\sigma^2 = y'My/\sigma^2 \sim \chi^2_{n-g} \quad \text{(Theorem 7(b), } r = g\text{)}$$
$$\text{SSR and SSE are independent } (AM = 0) \quad \text{(Lecture 10, Theorem 16(a))}$$
$$F = \frac{(\text{SSR}/\sigma^2)/(g-1)}{(\text{SSE}/\sigma^2)/(n-g)} \sim F_{g-1,n-g},\ \mathsf{P}(F > F_{g-1,n-g,1-\alpha}) = \alpha \quad \text{(F9; F10)}$$

□

The computations of Corollary 13 are arranged in the **analysis-of-variance table**: the degrees of freedom are those of the χ^2 distributions in the proof, and $n-1=(g-1)+(n-g)$ for SST; the **mean squares** are the sums of squares divided by them, and F is their ratio:

source	degrees of freedom	sum of squares	mean square
between groups	$g-1$	$\mathrm{SSR}=\sum_l n_l(\bar y_l-\bar y)^2$	$\mathrm{SSR}/(g-1)$
within groups	$n-g$	$\mathrm{SSE}=\sum_l\sum_{i\in\mathcal{I}_l}(y_i-\bar y_l)^2$	$s^2=\mathrm{SSE}/(n-g)$
total	$n-1$	$\mathrm{SST}=\sum_i(y_i-\bar y)^2$	

Example 14. In Lecture 11, Example 7, assume the normal linear model. Test $H_0\colon \alpha_1=\alpha_2=\alpha_3$ at level α.

Solution.

Group means $6,4,2$ (Lecture 11, Example 7); Lecture 11, Example 11 and Theorem 15; Theorem 12:

$$\bar y=28/6=14/3;\quad e=(-1,1,0,-1,1,0)',\ \mathrm{SSE}=4,\ r=\operatorname{rank}X=3,\ s^2=4/3$$

$$\mathrm{SSR}=3\Big(6-\frac{14}{3}\Big)^2+2\Big(4-\frac{14}{3}\Big)^2+\Big(2-\frac{14}{3}\Big)^2=\frac{48+8+64}{9}=\frac{40}{3},\qquad \mathrm{SST}=\frac{40}{3}+4=\frac{52}{3}$$

$\mathbf{1}_6=d_1+d_2+d_3$; Corollary 13 with $n=6>g=3$:

$$X\beta=\mu\mathbf{1}_6+\alpha_1d_1+\alpha_2d_2+\alpha_3d_3=X_1(\mu+\alpha_1,\ \mu+\alpha_2,\ \mu+\alpha_3)'\colon\ H_0\text{ is }\mu_1=\mu_2=\mu_3$$

$$F=\frac{(40/3)/2}{4/3}=5,\ \text{with 2 and 3 degrees of freedom: the test rejects } H_0\iff F_{2,3,1-\alpha}<5$$

source	degrees of freedom	sum of squares	mean square
between groups	2	40/3	20/3
within groups	3	4	4/3
total	5	52/3	

The same F by Theorem 11(a) in the design X ($r=3$): H_0 is $K\beta=0$, testable (the criterion of Example 3 for the rows); G, $\beta^0=(0,6,4,2)'$ as in Example 5:

$$K=\begin{pmatrix}0&1&-1&0\\0&0&1&-1\end{pmatrix}\colon\ -0+1-1+0=0,\ -0+0+1-1=0\colon\ \text{rows estimable}$$

$$c_1(0,1,-1,0)+c_2(0,0,1,-1)=(0,\ c_1,\ c_2-c_1,\ -c_2)=0\implies c_1=c_2=0\colon\ \operatorname{rank}K=2\quad\text{(F4)}$$

$$K\beta^0=(2,2)',\quad V=KGK'=\begin{pmatrix}\frac{5}{6}&-\frac{1}{2}\\-\frac{1}{2}&\frac{3}{2}\end{pmatrix}$$

$$ad-bc=\tfrac54-\tfrac14=1,\quad V^{-1}=\begin{pmatrix}\frac{3}{2}&\frac{1}{2}\\\frac{1}{2}&\frac{5}{6}\end{pmatrix}\qquad\text{(F3)}$$

$$(K\beta^0)'V^{-1}K\beta^0=4\Big(\tfrac32+\tfrac12+\tfrac12+\tfrac56\Big)=\tfrac{40}{3}=\mathrm{SSR},\qquad F=\frac{40/3}{2\cdot4/3}=5$$

Uncentered (Lecture 11, Theorem 17): $n\bar y^2=6\cdot\frac{196}{9}=\frac{392}{3}$, and $y'y=148=\frac{392}{3}+\frac{40}{3}+4$.

Practice. The Exercises and Problems of the textbook, Chapter 14: estimable functions and their best linear unbiased estimators, the normal linear model, t intervals and tests, testable hypotheses and the F test, and the one-way analysis of variance.